Ch. 15 CW #1

Principles of Microeconomics

December 3, 2025

Instructions

Please answer these questions on another sheet of paper and turn them in on Gradescope by next class.

Warm Up

1. Throwback to Ch. 4: What two characteristics make a market competitive?
2. Throwback to Ch. 14: What was our core assumption about firms? What do we assume they're trying to accomplish?

15.1. What Is a Competitive Market?

1. A perfectly competitive firm
   (a) chooses its price to maximize profits.
   (b) sets its price to undercut other firms selling similar products.
   (c) takes its price as given by market conditions.
   (d) picks the price that yields the largest market share.
2. Minnesota Milk is a thriving dairy company. It sells milk by the gallon, and the market price of a gallon is $6.
   (a) Why is it reasonable to assume that the milk market is competitive?
   (b) Please fill in the following table:

Quantity	Price	Total Revenue	Average Revenue	Marginal Revenue
1 gallon	$6			
2	6			
3	6			
4	6			
5	6			
6	6			
7	6			
8	6			

3. When a perfectly competitive firm increases the quantity it produces and sells by 10 percent, its marginal revenue _____ and its total revenue rises by _____.

   (a) falls; less than 10 percent

   (b) falls; exactly 10 percent

   (c) stays the same; less than 10 percent

   (d) stays the same; exactly 10 percent

15.2. Profit Maximization and the Competitive Firm's Supply Curve

1. Minnesota Milk is still thriving, and the market price of a gallon of milk is still $6.

   (a) Please fill in the following table.

Quantity	Total Revenue	Total Cost	Profit	Marginal Revenue	Marginal Cost	Change in Profit
0 gallons	$0	$3				
1	6	5				
2	12	8				
3	18	12				
4	24	17				
5	30	23				
6	36	30				
7	42	38				
8	48	47				

   (b) How many gallons should Minnesota Milk sell? There are two possible answers.

   (c) What is true of marginal revenue and marginal cost at the profit maximizing quantities?

(d) What is the relationship between marginal revenue and marginal cost when change in profit is positive? What about when it's negative?

(e) Keeping in mind your answer to (c), try to explain your answer to part (b).

2. A competitive firm maximizes profit by choosing the quantity at which

(a) average total cost is at its minimum.

(b) marginal cost equals the price.

(c) average total cost equals the price.

(d) marginal cost equals average total cost.